

Plan: **HymeKo-Gömb** (Hungarian for sphere) —
A Concentric Cascade of FIR Volume + HSiKAN Shell + CPML
Core
for Signed-Hypergraph Learning

2026-05-11

Etymology

HymeKo-Gömb: *gömb* is Hungarian for “sphere / ball / globe.” This name deliberately mirrors and complements **Nagare** (, the Japanese word for “flow”): the two sibling architectures of this codebase carry one Japanese and one Hungarian name, mediated by the HymeKo brand. Together they evoke a complete cosmology:

- **Nagare** (, Japanese, “flow”): the dataflow *runtime* (`hymeKo_nagare` crate, no-autograd, closed-form Clifford backward)
- **Gömb** (Hungarian, “sphere”): the concentric *architecture* (this plan; three nested computational shells)
- **HSMM** (Hajdu & Csapó 2026): the abstract *machine* that hosts both

Public-facing English alias: **HymeKo-Sphere**.

1 Scope

In contrast to the orthogonal-factorial design (where HSiKAN, CPML, Clifford-FIR are independently toggled axes), HymeKo-Gömb is a *cascade*: all three are mandatory but each operates at a distinct radius through the computation. The architecture is a single forward path with explicit hierarchical roles:

- **Outer shell (volume)**: M parallel Clifford-FIR filter banks process the same cycle pool with different learnable filter coefficients. Cheap, redundant coverage. Bio-analogue: V1 oriented-edge detectors — many parallel receptive-field copies at the cortical periphery.
- **Middle shell (single HSiKAN layer)**: one Catmull-Rom signed-spline aggregator refines the outer-shell output. Adds nonlinearity. Bio-analogue: V4 curvature / shape area — a single thick cortical layer transforming oriented primitives into shape representations.
- **Inner core (single CPML pass)**: $L = 3$ tier-stratified multi-layer aggregation compresses to the final per-vertex embedding. Bio-analogue: IT (inferotemporal cortex) — multi-scale integration producing object identity.

The architecture is loosely inspired by Feature Pyramid Networks (FPN, Lin et al. 2017), but rotated 90°: where FPN stacks resolution levels in DEPTH (lateral connections between levels), HymeKo-Gömb stacks specialisation levels at a SINGLE depth (the “radius” is the role-in-cascade, not the spatial scale).

2 Mathematical structure

Let $X \in \mathbb{R}^{|V| \times d_0}$ be the per-vertex learnable embedding. Let $\mathcal{C}$ be the cycle pool with associated edge-sign vectors. Let $\text{tier_of} : V \rightarrow \{0, \dots, L-1\}$ be the CPML tier assignment.

2.1 Outer shell (FIR volume)

Define M Clifford-FIR filters $\{(a^{(m)}, b^{(m)}) : m = 1, \dots, M\}$, each of length k . Per-cycle outer outputs:

$$\text{outer}_m[c, j] = \sum_{i=0}^{k-1} \text{proj}_{\sigma_i^{(c)}}(a_i^{(m)} + ib_i^{(m)}) \cdot X[v_i^{(c)}][j].$$

Per-vertex outer features via scatter-mean over the cycle pool:

$$H_{\text{outer}}^{(m)}[v][j] = \text{mean}_{c:v \in c} \text{outer}_m[c][j].$$

Concat over the M banks:

$$H_{\text{outer}}[v][\cdot] = \text{concat}_{m=1}^M H_{\text{outer}}^{(m)}[v][\cdot] \in \mathbb{R}^{M \cdot d_0}.$$

Optional softmax α_M -routing instead of concat for the multi-bank case — recovers the existing α_k routing primitive that drove Slashdot SOTA.

2.2 Middle shell (HSiKAN)

One Catmull-Rom signed-spline aggregation:

$$H_{\text{middle}} = \text{HSiKAN}(H_{\text{outer}}, \mathcal{C}, \sigma).$$

The spline aggregator is the existing `MixedAritySignedKAN` forward path with $L_{\text{HSiKAN}} = 1$ inner layer; sign branches via `use_minus_branch=True` as in the Slashdot recipe.

2.3 Inner core (CPML pass)

Tier-stratified $L = 3$ aggregation:

$$H_{\text{core}} = \text{CPML}_{L=3}(H_{\text{middle}}, \mathcal{C}, \text{tier_of}),$$

producing the final per-vertex embedding. Each CPML tier uses an MLP (the feasibility-stub) or one of $\{\text{HSiKAN}, \text{Clifford-FIR}\}$; default for HymeKo-Gömb is MLP (since the spheres already host the specialised aggregators).

2.4 Edge predictor head

$$s(u, v) = \text{MLP}_{\text{head}}(\text{concat}(H_{\text{core}}[u], H_{\text{core}}[v])).$$

2.5 Parameter count

- Outer shell: $M \cdot (2k + d_0)$ params (Clifford coefs + bias per bank). For $M = 8, k = 3, d_0 = 32$: $8 \cdot 38 = 304$.
- Middle shell: existing HSiKAN at $M \cdot d_0 \rightarrow d_0$ width: typically $\sim 10^4$ for $d_0 = 32, \text{grid} = 5$.
- Inner core: existing CPML at $d_0 \rightarrow d_0$ over $L = 3$ tiers, $\sim 10^4$ params for MLP-stub tiers.
- Head: $2d_0 \cdot d_{\text{head}} + d_{\text{head}}$, $\sim 10^3$.
- Plus per-vertex embedding: $|V| \cdot d_0 \cdot 4$ bytes = 4.2 MB for Epinions at $d_0 = 32$.

Total: ~ 4.3 MB on Epinions. Comparable to the existing `edge_cr 5-seed-SOTA` recipe.

3 Comparison to the factorial

The factorial says *which subset of $\{A, C, F\}$ matters*. HymeKo-Gömb says *all three matter in this specific order*. The two designs are complementary:

- The factorial gives *attribution* (which axis carries signal).
- HymeKo-Gömb gives a *single high-performance architecture* that integrates all three.

If the factorial reveals that all three single-axis effects $\alpha, \gamma, \varphi > 0$ with low interaction terms, HymeKo-Gömb should outperform any single-axis cell — it gets the sum of the three main effects. If interactions are strong (e.g., $\alpha\gamma\varphi > 0$), HymeKo-Gömb gets that bonus too.

If the factorial shows null effects on a given dataset, HymeKo-Gömb is the cheap test of the cascade hypothesis: maybe the THREE-ARRANGEMENT is necessary even when no SINGLE axis is.

4 Affected files

signedkan_wip/src/ hymeko_kyu.py KyuConfig OuterFIRShell MiddleHSiKAN InnerCPMLCore HymeKoKyu run_kyu_sweep.py tests/test_kyu.py	NEW: Sphere architecture (M, k, L_cpml, sphere widths) M parallel Clifford-FIRs wrapped Catmull-Rom tier-stratified MLP/CPML the full cascade NEW: M / L_cpml sweep runner NEW: forward + backward shapes, no-cycles ablation, inner-tier reachability test
docs/plans/2026-05-11-hymeko-gomb-sphere/ plan.tex plan.pdf plan.tikz plan.mmd	this doc built sphere-shell diagram cascade flow

CORE.YAML items touched: none. Sphere is an additive new model class; existing pipelines unmodified.

5 Test strategy

Unit tests in test_kyu.py:

- **outer_shell_M_banks_independent:** M banks produce M -way concat with distinct coefficient initialisations.
- **middle_layer_nonlinearity:** passing identity features through Outer + Middle produces non-identity output (HSiKAN spline activates).
- **inner_core_tier_reachability:** outer features reach the innermost tier through the CPML inward funnelling (reuses the CPML feasibility test pattern).
- **forward_backward_shape:** end-to-end on a 30-vertex fixture, no NaN gradients.
- **no_cycles_ablation:** cycles= $\emptyset$ all three shells degenerate to learnable-embedding pass-through, output is the embedding-floor predictor.

- **parameter_count_within_budget:** at $M = 8, k = 3, d_0 = 32, |V| = 10^5$, total params 5 MB.

Real-data smoke: train on Bitcoin OTC 50 epochs single seed, verify val AUC ≥ 0.85 (better than the factorial’s mean-MLP cell baseline).

Validation gate: HymeKo-Gömb is reported as a clean architectural contribution only if its 5-seed paired AUC on at least one dataset is $\geq$ both:

1. the existing edge_cr Slashdot SOTA recipe at iso-parameters,
2. the best-cell of the orthogonal factorial.

If neither, it’s an interesting null result documented as “cascade arrangement does not exceed best-axis-only design.”

6 Performance budget

- Forward latency at Epinions scale ($|V| = 131K, |C| = 530K, M = 8, d_0 = 32$): expected ~ 200 ms (outer shell is cheap by virtue of FIR primitives; middle and inner are existing operators).
- Peak RSS: ≤ 5 GB. Outer shell’s M -way concat raises per-vertex feature width by a factor of M ; mitigated by setting $M = 4$ if 5GB is breached.
- Per-epoch training time (Epinions, 50 epochs): expected ≤ 3 min, comparable to the kitchen-sink and edge_cr recipes.

7 Risk anticipation

- **Outer-shell collapse:** if M FIR banks converge to similar coefficients, the outer shell adds no diversity over a single Clifford-FIR layer. Mitigation: encourage diversity via orthogonality regularisation on the bank coefficients, or diverse initialisation (varied seeds per bank).
- **Middle-layer redundancy:** if outer + inner together already capture everything, HSiKAN middle adds no signal. Detected via the no-HSiKAN-middle ablation cell. Mitigation: if HSiKAN-middle is null, drop it from the published architecture and report “2-shell suffices.”
- **Inner-core dominance:** if CPML core does all the work (because it has the most parameters), the outer and middle shells are spectators. Detected via the corresponding ablations. Mitigation: published architecture clearly attributes work via the ablation.
- **M -way concat memory:** M banks $\times d_0$ per vertex $\times |V|$ floats = $M \cdot d_0 \cdot 4|V|$ bytes. At $M = 8, d_0 = 32, |V| = 131K$ that’s ~ 130 MB; fine. At $M = 32$ it’s ~ 540 MB; still fine. At $M = 128$ it’s 2 GB; bound $M \leq 64$ as a default policy.

8 Rollback path

HymeKo-Gömb is a single Python module + a unit-test file. Removing it removes the architecture; no existing pipeline regresses. Drop the plan dir + the module + the test file = clean rollback.

9 Sequencing

1. Implement OuterFIRShell (re-uses ClifFIRTierAggregator per bank).
2. Implement MiddleHSiKAN (wraps existing MixedAritySignedKAN).
3. Implement InnerCPMLCore (re-uses existing CPML).
4. Compose: HymeKoKyu(outer, middle, inner, head).

5. Unit tests (all 6 above).
6. Smoke train on Bitcoin OTC seed 0 (~ 5 min).
7. Ablations: drop one shell at a time, measure per-cell paired on 5 seeds. This is the cascade-attribution analogue of the factorial’s main-effects decomposition.

10 Venue

If HymeKo-Gömb beats the factorial’s best cell by ≥ 1.5 pp paired on at least one dataset:

Primary: ICLR 2027 (single clear architectural contribution with cascade decomposition and biological motivation; “concentric-sphere signed hypergraph learning”).

Secondary: NeurIPS GLF workshop (if the lift is borderline).

If null: archived as a documented negative result alongside the factorial; the factorial’s best-cell becomes the recommended architecture and HymeKo-Gömb is footnoted as “cascade does not exceed orthogonal-best.”

11 Connection to the cosmology

The three architectural plans of 2026-05-11 form a coherent set:

- 2026-05-11-hsikan-cpml-fir-orthogonal/: the **factorial** — a tool for architectural attribution via orthogonal decomposition.
- 2026-05-11-hymeko-gomb-sphere/: this plan — the **cascade** architecture that consumes the factorial’s components and arranges them concentrically.
- 2026-05-11-hymeko-nagare-flow/: **Nagare** — the runtime / framework that hosts both, with closed-form Clifford backward and no autograd graph.

Plus the critical-frozen **CRITICAL-nagare-to-hsmm-compiler/** that bridges Nagare to the HSMM substrate when the FPGA prerequisite lands.

Together: **factorial** *measures*, **sphere** *combines*, **flow** *executes*, **HSMM** *hosts*. Four plan dirs, one research program.